

Aleena Varghese

AI Engineer | Machine Learning Engineer | Data Scientist

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| portfolio-lake-one-10.vercel.app

Professional Summary

AI Engineer with hands-on experience developing production-grade Generative AI and Retrieval-Augmented Generation (RAG) applications for enterprise knowledge retrieval. M.Sc. Computer Science (Data Analytics) graduate with expertise in Python, Machine Learning, Deep Learning, Large Language Models (LLMs), Computer Vision, FastAPI, Django, React, and Next.js. Experienced in designing intelligent, scalable AI solutions that solve real-world business challenges through data-driven decision-making.

Experience

AI Engineer Intern

Thiruvananthapuram, Kerala (Remote)

March 2026 – June 2026

Lightrains Technolabs

- Engineered a production-grade multimodal AI diagnostic assistant using Python, FastAPI, and Next.js that enabled users to retrieve accurate information from technical manuals through intelligent AI-powered question answering.
- Built a Retrieval-Augmented Generation (RAG) pipeline integrating knowledge graphs, vector search, and document parsing to improve the accuracy and relevance of information retrieval from unstructured technical documents.
- Implemented Vision Language Models (VLMs) for semantic image retrieval and visual grounding, enabling context-aware understanding of technical diagrams and improving AI-assisted diagnostics.
- Designed AI safety guardrails, contextual memory, confidence validation, and escalation workflows to improve the reliability, explainability, and safety of AI-generated responses.

Projects

HR Assistant System | Django, React, TypeScript, Llama 3.3, SQLite

- Developed an AI-powered HR Assistant using Django, React, TypeScript, and Llama 3.3 to automate employee queries related to payroll, leave management, company policies, and HR documentation.
- Developed an interactive AI chatbot interface integrating LLM workflows to deliver secure, contextual, and policy-aware HR assistance.
- Implemented role-based access control, audit logging, and deterministic processing to improve transparency, security, and consistency in HR operations.

SignBot – Sign Language Recognition | Python, OpenAI CLIP, SVM, Google Gemini AI, Streamlit

- Developed a real-time Sign Language Recognition system using OpenAI CLIP for feature extraction and an SVM classifier for gesture recognition.
- Integrated Google Gemini AI and Text-to-Speech (TTS) to generate context-aware textual and voice responses, improving accessibility for users with hearing and speech impairments.
- Built an end-to-end AI pipeline covering dataset preparation, feature extraction, model training, evaluation, and real-time inference.

Age, Gender, and Emotion Prediction | Python, TensorFlow, Keras, OpenCV

- Developed CNN-based deep learning models using TensorFlow/Keras to predict age, gender, and facial emotions from images using the UTKFace and FER2013 datasets.
- Built complete data preprocessing, training, evaluation, and inference pipelines using TensorFlow, Keras, OpenCV, NumPy, Pandas, and Scikit-learn.
- Evaluated model performance using standard machine learning metrics to ensure reliable facial attribute prediction.

Education

M.Sc. Computer Science (Data Analytics)

Digital University Kerala

CGPA: 9.12

Thiruvananthapuram, Kerala

2024 – 2026

B.Sc. Mathematics

Mar Ivanios College (Autonomous)

CGPA: 7.71

Thiruvananthapuram, Kerala

2021 – 2024

Technical Skills

Programming Languages: Python, SQL, JavaScript

Artificial Intelligence: Deep Learning, Machine Learning, Generative AI, Large Language Models, Retrieval-Augmented Generation (RAG), Computer Vision, Natural Language Processing (NLP), Convolutional Neural Networks (CNN), Support Vector Machines (SVM), Prompt Engineering

Frameworks & Libraries: FastAPI, Django, React, Next.js, Streamlit, TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib

Tools: Git, GitHub, OpenAI API, Google Gemini AI, LightRAG, MinerU, CLIP, Vision Language Models (VLMs), PydanticAI, LangChain, Docker

Databases: SQLite, PostgreSQL, Vector Databases, Knowledge Graph Stores

Professional Skills

Analytical Thinking • Problem Solving • Communication • Technical Presentations • Team Collaboration • Technical Documentation • Adaptability • Continuous Learning

Certifications

- Introduction to Large Language Models — Google Cloud
- Data Analytics Track — Google Cloud Career Launchpad
- Computing Foundations Track — Google Cloud Career Launchpad
- Python Data Structures — University of Michigan via Coursera
- Introduction to Git and GitHub — Google via Coursera
- Data Analytics Job Simulation — Deloitte
- Data Visualization — Tata Group